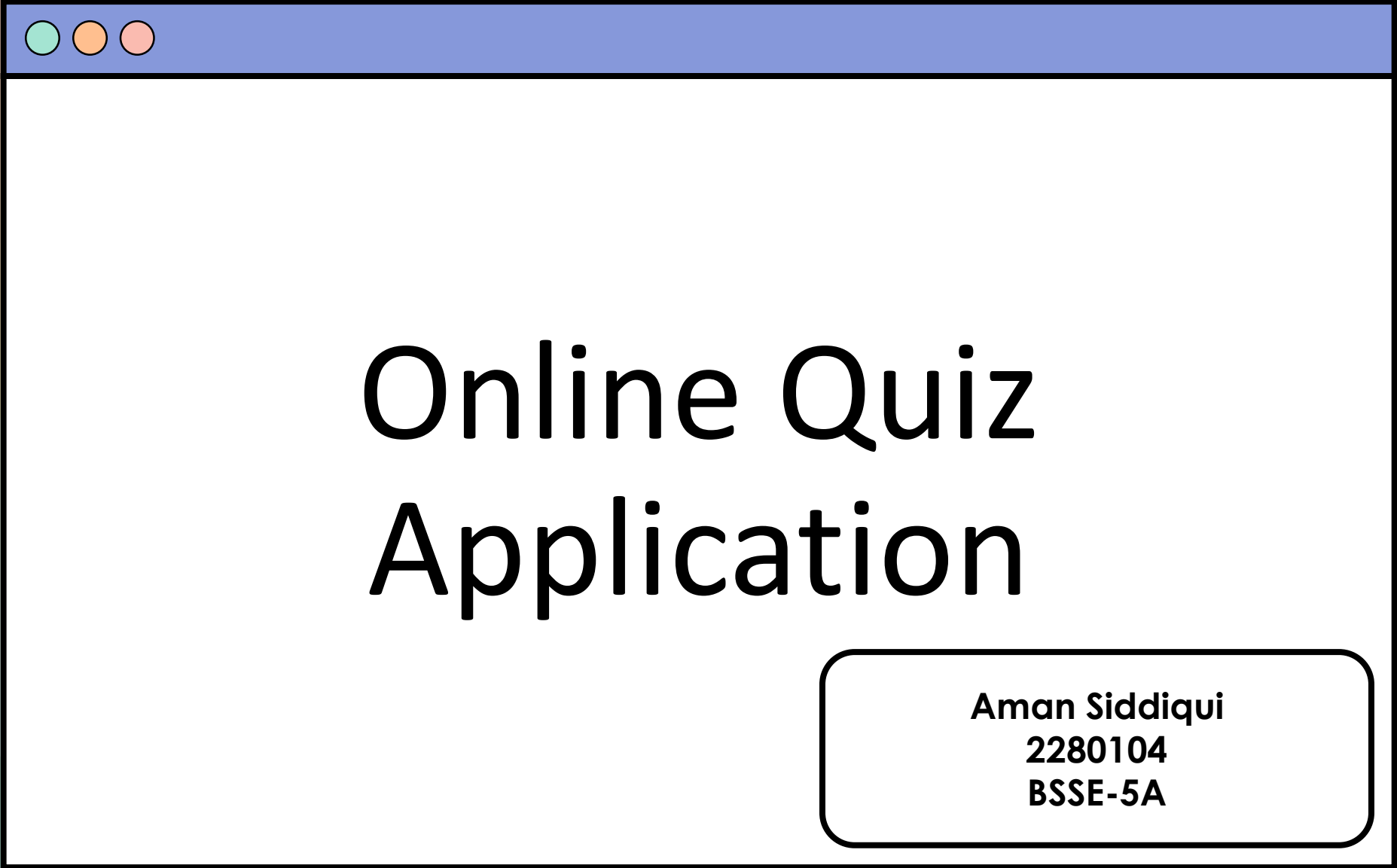




Online Quiz Application

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Introduction

The Online Quiz Application is a cutting-edge platform designed to make learning interactive, efficient, and enjoyable. With a wide range of quiz categories, adjustable difficulty levels, and real-time scoring, it caters to users of all ages and expertise. Built using Java, this application combines secure authentication, seamless user experience, and detailed progress tracking to help users test their knowledge and grow their skills effectively. Perfect for students the Online Quiz Application transforms the way knowledge is assessed and retained.

Features

- User Registration & Authentication
- User Role
- Quiz Categories
- Difficulty Levels
- Quiz Rules
- Scoring
- Result Display
- Correct Answers (if any)
- Quiz History
- Reattempt Quiz

START



12/27/2024

Inheritance

- Inheritance enables a class to inherit methods and properties from another class.
- The `RegularUser` class inherits from the `User` class, allowing it to use the `registrationNumber` and `password` fields.

```
class RegularUser extends User {  
    private List<String> quizHistory = new ArrayList<>();  
  
    public RegularUser(String registrationNumber, String password) {  
        super(registrationNumber, password);  
    }  
}
```

Abstraction

- Abstraction hides implementation details and shows only essential features.
- The User class is abstract, which means it cannot be instantiated directly and provides a base for other classes like RegularUser.

```
abstract class User {  
    protected String registrationNumber;  
    protected String password;
```

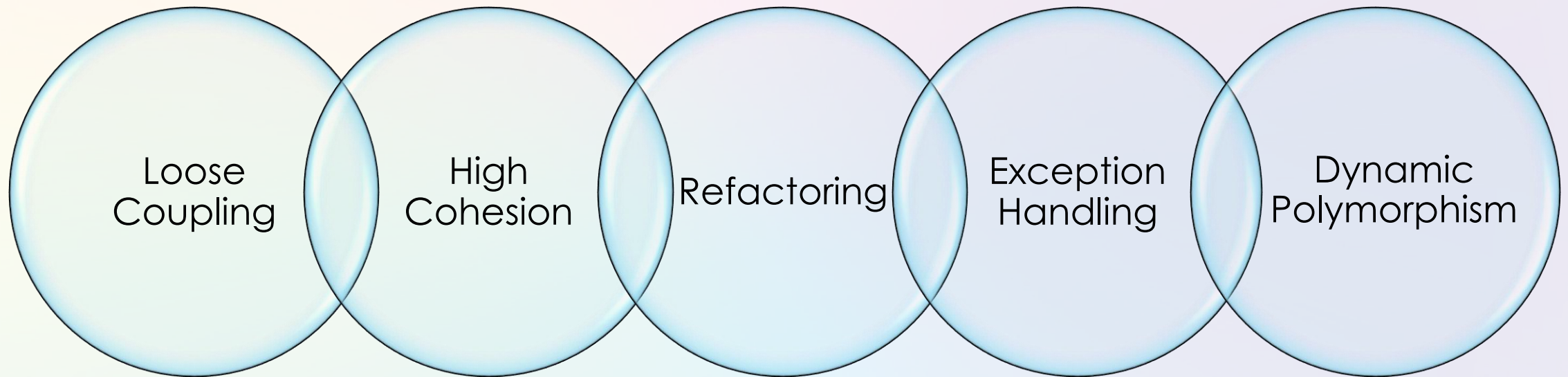
```
public abstract void displayRole();
```

Encapsulation

- Encapsulation is the concept of wrapping data and methods into a single unit.
- This code uses private fields and public methods to ensure proper access and modification.

```
protected String registrationNumber;  
protected String password;
```

Concepts After Mids



START



Loose Coupling

- Loose coupling means that components are independent of each other.
- The code structure allows different components (like RegularUser and Quiz) to interact without tightly depending on each other.

```
Quiz quiz = new Quiz(subject, difficultyLevel); // Loose coupling between Quiz and RegularUser  
generateQuestions(quiz);
```

High Cohesion

- High cohesion is maintained by ensuring that each class has a single responsibility.
- The Quiz class manages only quiz-related functionality, and the User class handles user-related actions.

```
class Quiz {  
    private String category;  
    private String difficultyLevel;  
    private List<String> questions;  
    private List<String[]> options;  
    private List<String> correctAnswers;  
}
```

Refactoring

- The code can be refactored to improve readability and maintainability.
- This code has sections where the flow could be simplified or improved, but it's already modular and clean.

```
private static void generateQuestions(Quiz quiz) {
    String subject = quiz.getCategory();
    try {
        switch (subject) {
            case "Math":
                if (quiz.getDifficultyLevel().equals("Easy")) {
                    quiz.addQuestion("What is 5 + 3?", new String[]{"6", "7", "8", "9"}, "C");
                    quiz.addQuestion("Solve: 10 - 2.", new String[]{"8", "7", "3", "9"}, "A");
                    quiz.addQuestion("What is 4 multiplied by 2?", new String[]{"6", "8", "10", "12"}, "B");
                    quiz.addQuestion("Divide 20 by 5.", new String[]{"2", "3", "4", "5"}, "C");
                    quiz.addQuestion("What is the square of 3?", new String[]{"6", "7", "8", "9"}, "D");
                } else if (quiz.getDifficultyLevel().equals("Medium")) {
                    quiz.addQuestion("What is 15% of 200?", new String[]{"25", "30", "35", "40"}, "B");
                    quiz.addQuestion("Solve: 12 × (4 + 2).", new String[]{"60", "72", "48", "24"}, "B");
                    quiz.addQuestion("What is the square root of 81?", new String[]{"7", "8", "9", "10"}, "C");
                    quiz.addQuestion("If x = 7, find the value of 3x + 2.", new String[]{"23", "24", "25", "26"}, "A");
                    quiz.addQuestion("Simplify: (5/4) + (3/4).", new String[]{"1", "2", "3", "4"}, "B");
                } else if (quiz.getDifficultyLevel().equals("Hard")) {
                    quiz.addQuestion("What is 25 × 24?", new String[]{"600", "575", "625", "480"}, "A");
                    quiz.addQuestion("Solve: (18 × 3) - (12 ÷ 2).", new String[]{"48", "45", "51", "43"}, "A");
                    quiz.addQuestion("Find the value of x in 3x + 5 = 20.", new String[]{"5", "7", "6", "8"}, "A");
                    quiz.addQuestion("What is the value of 103?", new String[]{"100", "1000", "10000", "100000"}, "B");
                    quiz.addQuestion("Simplify: (7 × 3) - (2 × 4).", new String[]{"13", "17", "19", "20"}, "A");
                }
            }
        }
    } catch (Exception e) {
        break;
    }
}
```

Dynamic Polymorphism

- Dynamic polymorphism occurs when a method call is resolved at runtime.
- It is demonstrated in the code through method overriding (e.g., `displayRole()` in the User class).

```
public abstract void displayRole();
```

```
@Override  
public void displayRole() {  
    System.out.println("Role: Regular User");  
}
```

```
user.displayRole();
```

Exception Handling

- Exception handling is implemented to catch invalid inputs, such as incorrect registration number formats or out-of-bound subject choices.
- For example, try-catch blocks are used around registration number and password validation to provide meaningful error messages.

```
try {
    for (int i = 0; i < quiz.getQuestions().size(); i++) {
        System.out.println("\n\n" + (i + 1) + ". " + quiz.getQuestions().get(i));

        // Defensive programming for options retrieval
        if (quiz.getOptions(i) == null || quiz.getOptions(i).length == 0) {
            System.out.println("Error: No options available for this question.");
            continue; // Skip this question
        }

        String[] opts = quiz.getOptions(i);
        for (int j = 0; j < opts.length; j++) {
            System.out.println("    " + (char) ('A' + j) + ". " + opts[j]);
        }

        System.out.print("Your Answer (A/B/C/D): ");
        String answer = sc.nextLine().toUpperCase();

        // Validate user input
        if (!answer.matches("[A-D]")) {
            System.out.println("Invalid input. Please enter A, B, C, or D.");
            i--; // Repeat the question
            continue;
        }

        String correctAnswer = quiz.getCorrectAnswer(i);
        if (correctAnswer == null) {
            System.out.println("Error: No correct answer found for this question.");
            continue; // Skip this question
        }

        if (answer.equals(correctAnswer)) {
            score++;
        } else {
            incorrectAnswers.add((i + 1) + ". " + quiz.getQuestions().get(i));
            incorrectAnswers.add("Correct Answer: " + correctAnswer);
        }
    }
} catch (IndexOutOfBoundsException e) {
    System.out.println("Error: Question or answer index is out of bounds. " + e.getMessage());
} catch (Exception e) {
    System.out.println("An unexpected error occurred during the quiz: " + e.getMessage());
}
```

Output

```
OnlineQuiz [Java Application] C:\Users\The Laptop Hut\.p2\pool\plugins\org.eclipse.justj.open
Welcome to the Online Quiz System
Enter Registration Number (8 digits): 123456789
Invalid Registration Number. Please enter exactly 8 digits.
12345678
Enter Password (5 digits): 123456
Invalid Password. Please enter exactly 5 digits.
12345
Role: Regular User

Available Subjects

1. 📖 Math
2. 🔬 Science
3. 📚 History
4. 🌐 Geography
5. 📖 Literature
6. 🧠 General Knowledge
Choose a subject (1-6): 1

Choose Difficulty Level ⚡

1. ☼ Easy - Take it light and simple!
2. ⚡ Medium - A balanced challenge awaits.
3. 🏔 Hard - Only for the brave!
Enter Difficulty Level (1-3): 1
You selected Easy!
```

```

=====
***          QUIZ RULES          ***
=====
🕒 1. You only have 2 minutes to answer all questions.
⌚ 2. You have to select an option before you can go to the next question.
🔄 3. There's no going back after selecting an option.
🏆 4. At the end of the quiz, your score will be displayed.
🍀 Good luck, and give your best shot! 🍀

=====

Press Enter to start the quiz...

Quiz Starting...

1. What is 5 + 3?
A. 6
B. 7
C. 8
D. 9
Your Answer (A/B/C/D): c

2. Solve: 10 - 2.
A. 8
B. 7
C. 3
D. 9
Your Answer (A/B/C/D): a

```

Output

3. What is 4 multiplied by 2?

- A. 6
- B. 8
- C. 10
- D. 12

Your Answer (A/B/C/D): b

4. Divide 20 by 5.

- A. 2
- B. 3
- C. 4
- D. 5

Your Answer (A/B/C/D): c

5. What is the square of 3?

- A. 6
- B. 7
- C. 8
- D. 9

Your Answer (A/B/C/D): d

Quiz Completed!

Your Score: 5/5

Quiz History: [Math Quiz (Easy)]

All answers were correct!

Do you want to continue to another quiz? (yes/no): yes

Available Subjects

1. 📖 Math
2. 🔬 Science
3. 📜 History
4. 🌐 Geography
5. 📖 Literature
6. 🧠 General Knowledge

Choose a subject (1-6): 2

⚡ Choose Difficulty Level ⚡

1. 🌟 Easy - Take it light and simple!
2. ⚡ Medium - A balanced challenge awaits.
3. 🧛 Hard - Only for the brave!

Enter Difficulty Level (1-3): 2

You selected Medium!

**** QUIZ RULES ****

- 🕒 1. You only have 2 minutes to answer all questions.
- 🕒 2. You have to select an option before you can go to the next question.

Output

3. There's no going back after selecting an option.
4. At the end of the quiz, your score will be displayed.
Good luck, and give your best shot!

=====

Press Enter to start the quiz...

Quiz Starting...

1. What is the atomic number of carbon?

- A. 6
- B. 12
- C. 8
- D. 10

Your Answer (A/B/C/D): a

2. Which organelle is responsible for energy production in cells?

- A. Nucleus
- B. Mitochondria
- C. Endoplasmic Reticulum
- D. Golgi Apparatus

Your Answer (A/B/C/D): b

3. What is the chemical formula for methane?

- A. CO₂
- B. H₂O
- C. CH₄
- D. CO

Your Answer (A/B/C/D): d

4. Which gas is responsible for global warming?

- A. Oxygen
- B. Nitrogen
- C. Carbon Dioxide
- D. Hydrogen

Your Answer (A/B/C/D): d

5. What is the boiling point of nitrogen?

- A. -180°C
- B. -196°C
- C. -150°C
- D. -210°C

Your Answer (A/B/C/D): c

Quiz Completed!

Your Score: 2/5

Quiz History: [Math Quiz (Easy), Science Quiz (Medium)]

Incorrect Answers:

3. What is the chemical formula for methane?

Correct Answer: C

4. Which gas is responsible for global warming?

Correct Answer: C

5. What is the boiling point of nitrogen?

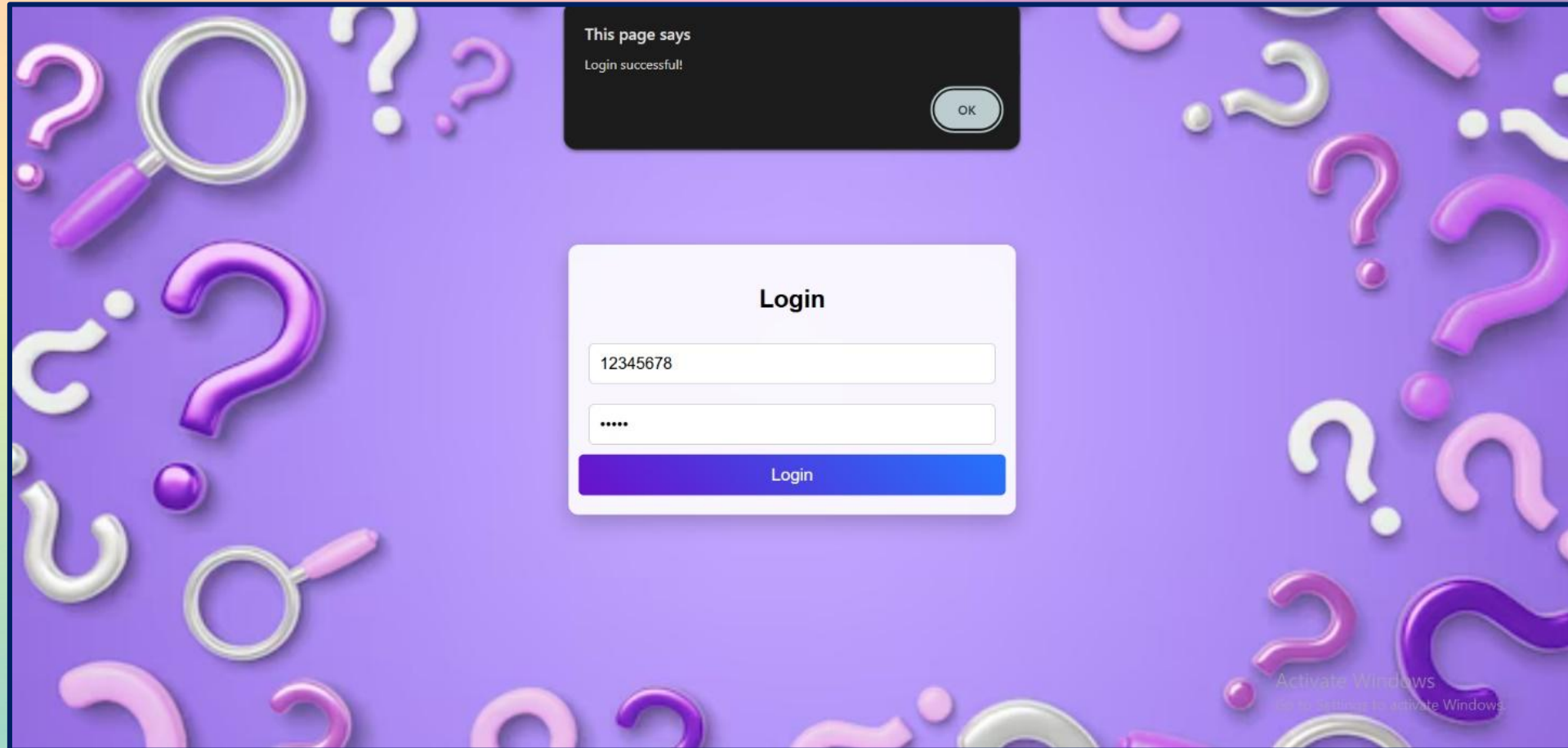
Correct Answer: B

Do you want to continue to another quiz? (yes/no): no

Thank you for taking the quiz! Goodbye.

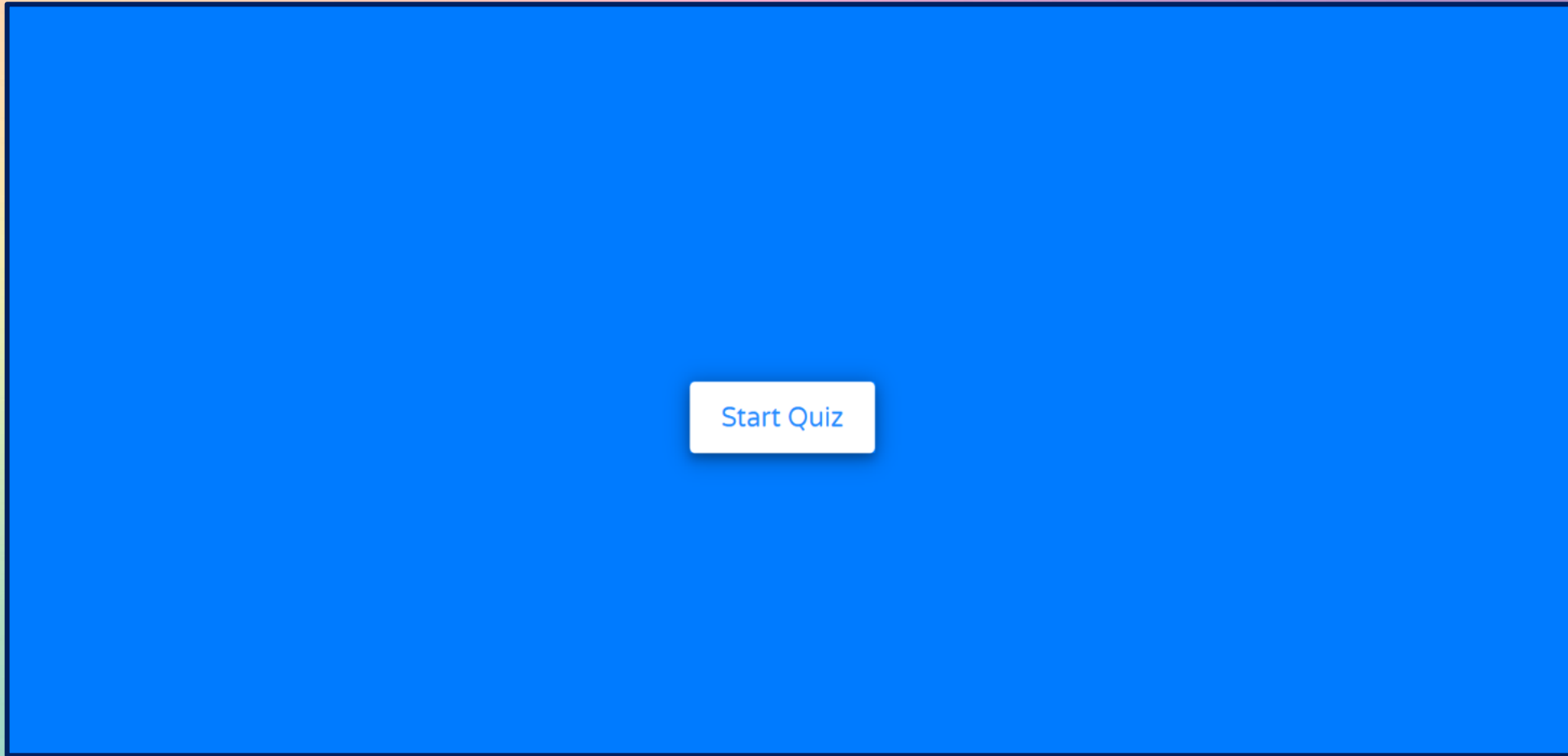
Graphical User Interface

❑ Login



Graphical User Interface

☐ Start Quiz



START



12/27/2024

17

Graphical User Interface

☐ Subject Selection

Select a Subject

Math

Science

History

Geography

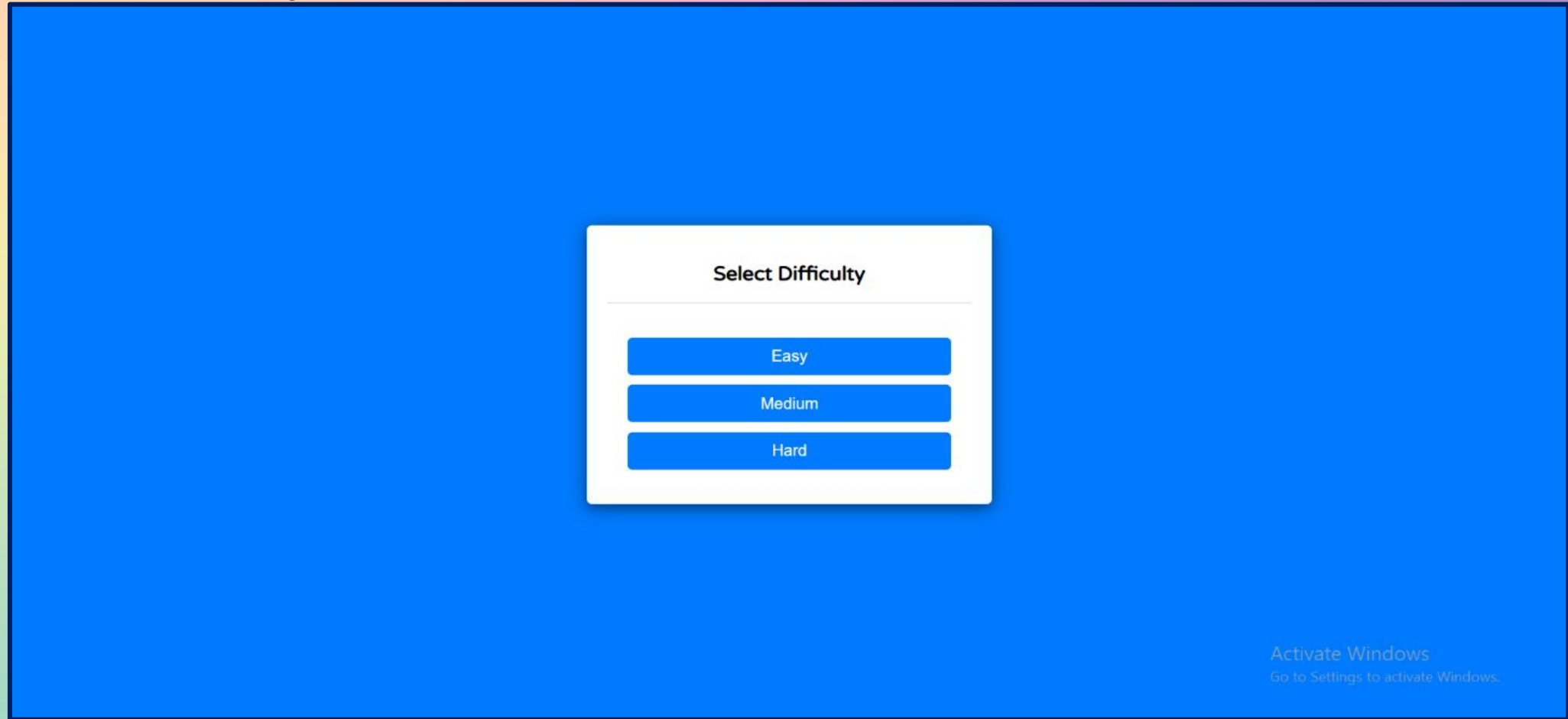
Literature

General Knowledge

Activate Windows
Go to Settings to activate Windows.

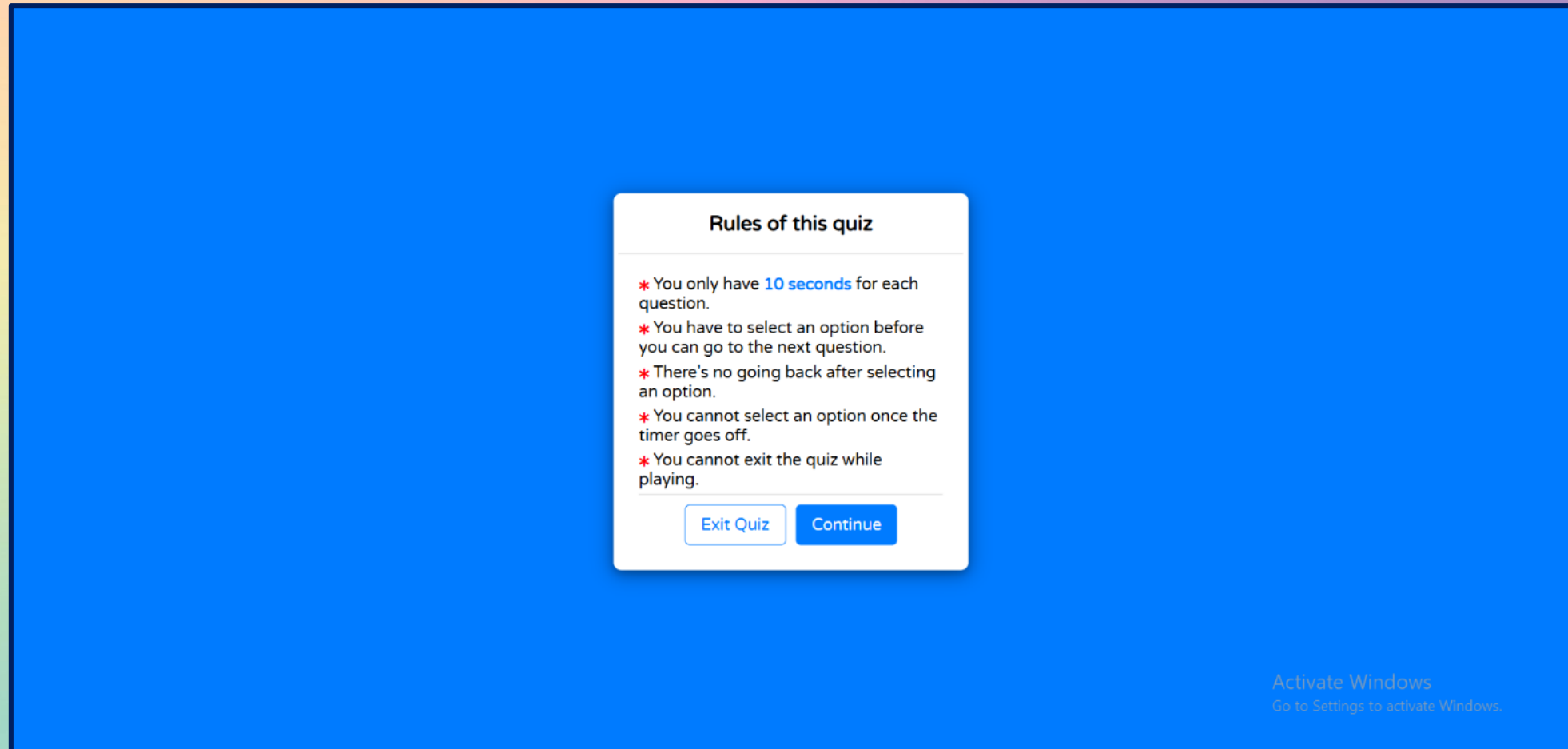
Graphical User Interface

❑ Difficulty Selection



Graphical User Interface

❑ Rules Of Quiz



START

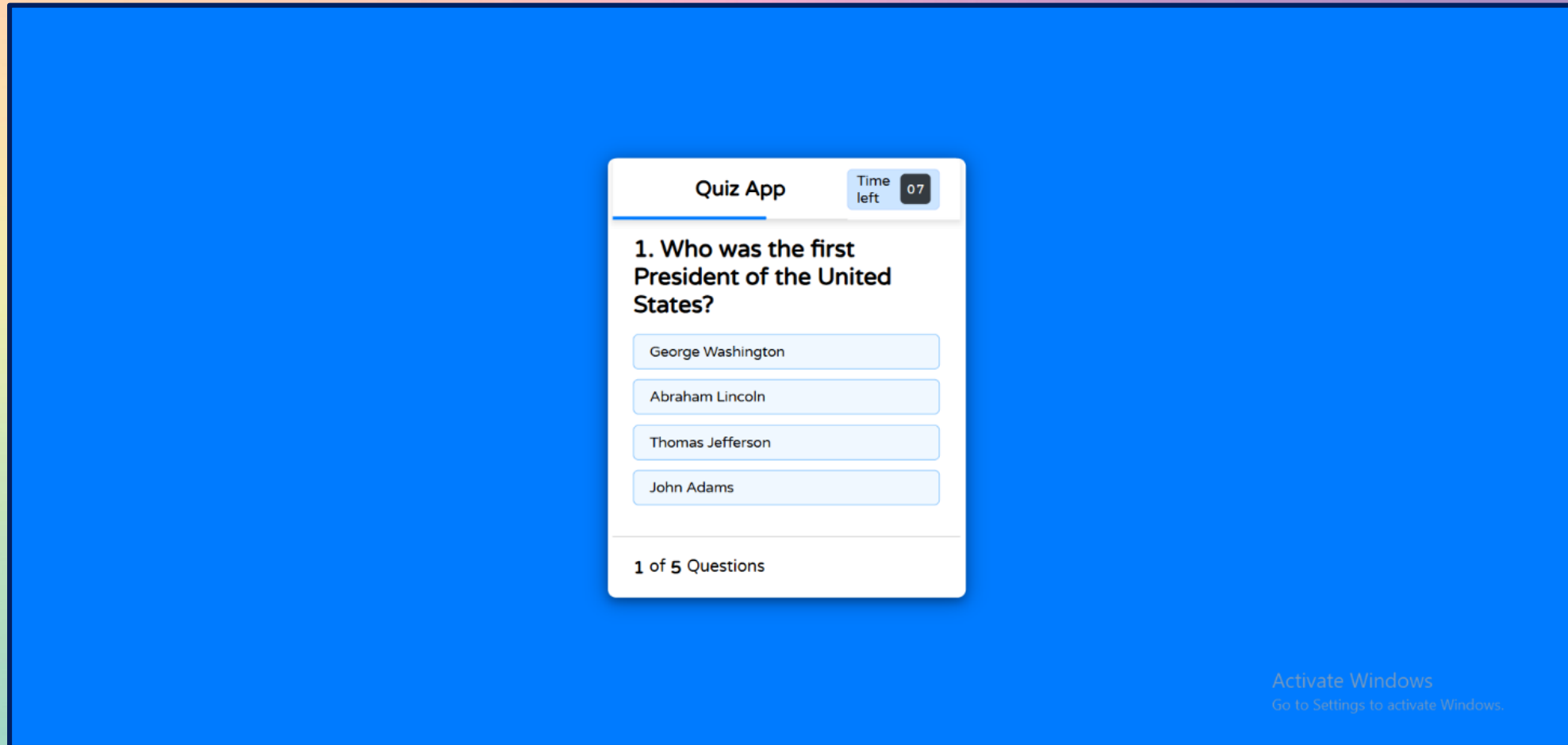


12/27/2024

20

Graphical User Interface

❑ Questions



START

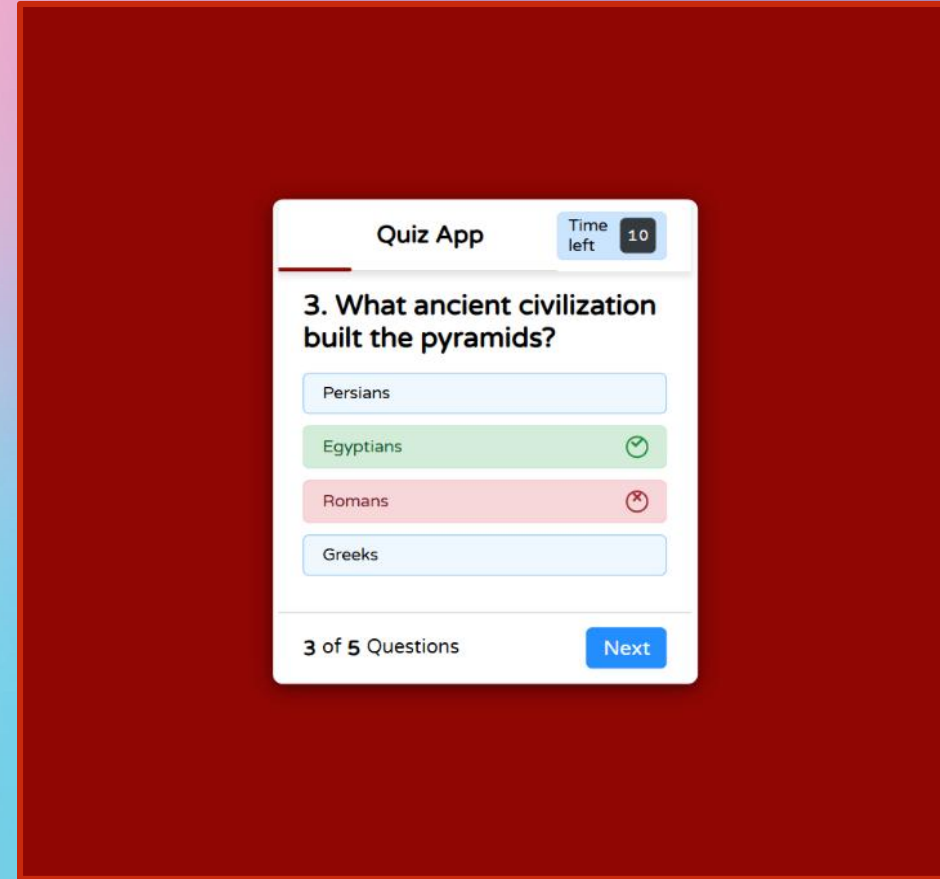
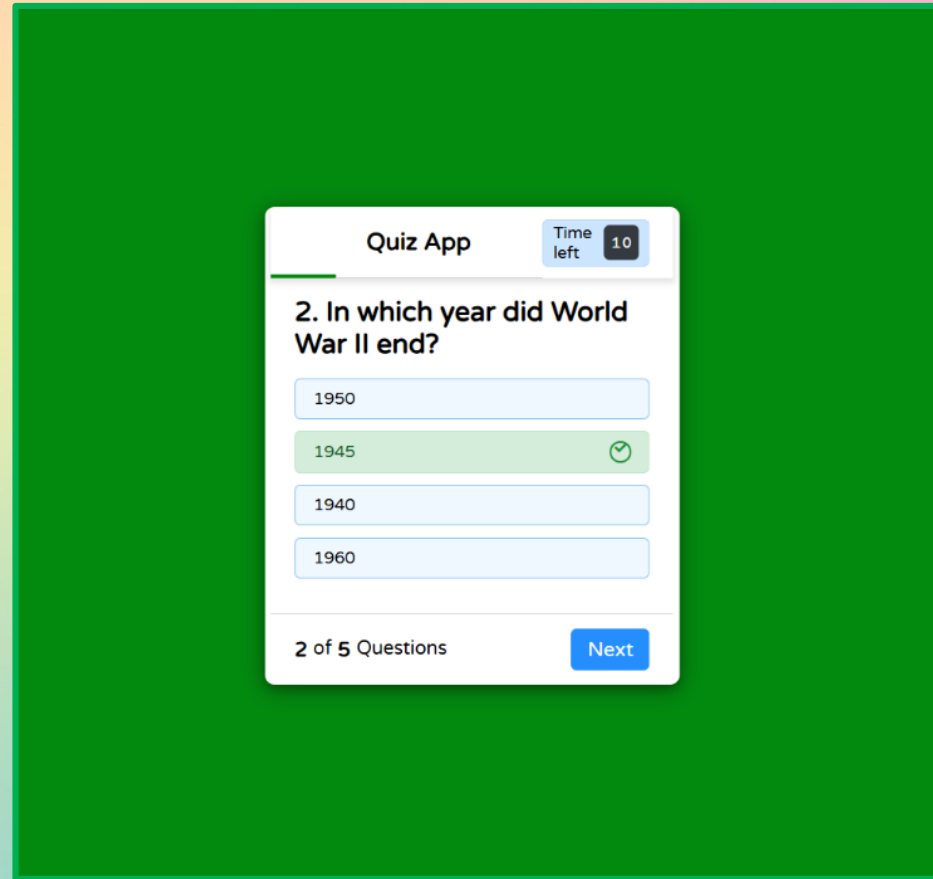


12/27/2024

21

Graphical User Interface

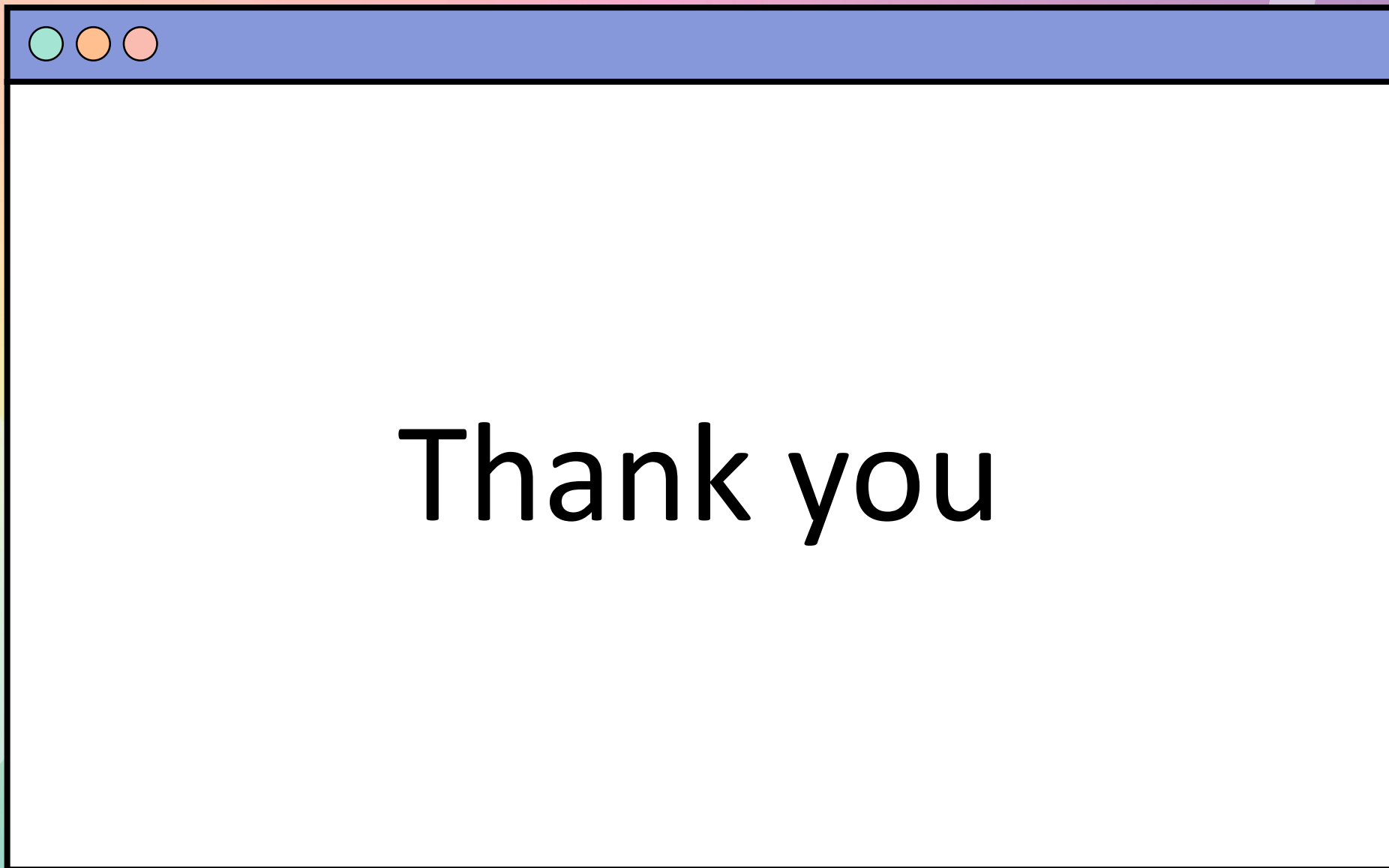
❑ Right/Wrong Answer



Graphical User Interface

❑ Quiz Completion





Thank you